

Experiment #(9)

Galvanometer

الجلفانوميتر



Objective

To convert the galvanometer into Ammeter and Voltmeter.

Apparatus

Power Supply



Galvanometer



Ammeter



Voltmeter



Banana Wires



Resistance Box



Rheostat



Theory

Part (1): CONVERSION OF GALVANOMETER INTO AMMETER

To convert a Galvanometer into an Ammeter, a very low resistance known as "shunt" resistance is connected in parallel to Galvanometer.

$$I_{\max} = I_{G,\max} + I_S$$

$$V_S = V_G$$

$$I_S R_S = I_{G,\max} R_G$$

$$(I_{\max} - I_{G,\max}) R_S = I_{G,\max} R_G$$

$$R_S = \frac{I_{G,\max} R_G}{I_{\max} - I_{G,\max}} \dots\dots\dots(1)$$

$$L = \frac{R_S}{\lambda} \times 10^5 \text{ cm} \dots\dots\dots(2)$$

$$\lambda = 180 \, \Omega/\text{Km}, I_{G,\max} = 0.5\text{mA}$$

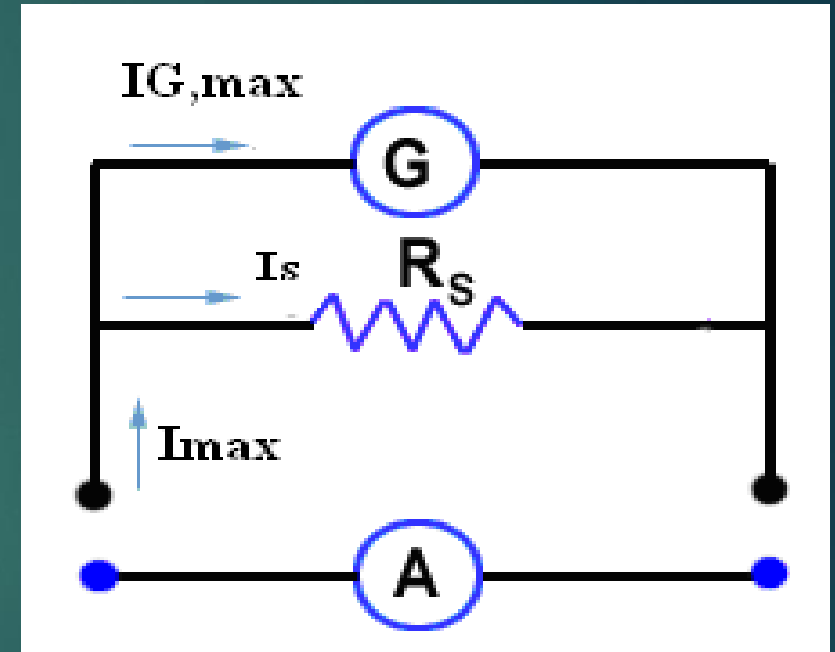
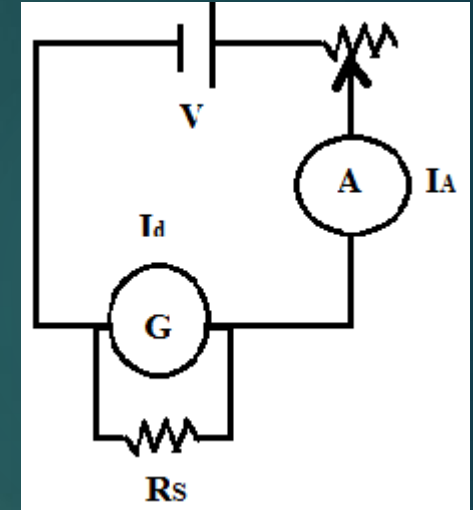


Figure 1

Procedure

1. Calculate the value of the shunt of resistance (R_s) from relation (1) to build up an ammeter measuring a maximum current of $I_{\max} = 1\text{Amp}$.
2. Calculate the value of the length of the wire that equivalent to R_s from relation (2).
3. Connect the shunt of resistance (R_s) to the galvanometer as shown in figure 1.
4. Connect the circuit shown in the Figure 2
5. By the rheostat change the current in the circuit and record the reading of the ammeter (I_A) and the built ammeter (I_d) and record the readings in table 1.
6. Draw the relation between the currents I_A and I_d .
7. From the slope calculate the efficiency of the built ammeter



Figure(2)

Part (2): CONVERSION OF GALVANOMETER INTO VOLTMETER

To convert a Galvanometer into voltmeter, a very high resistance R_x known as "series resistance" is connected in series with the galvanometer.

$$V = I_G (R_G + R_X)$$

$$V_{\max} = I_{G,\max} (R_G + R_X)$$

$$R_X = \frac{V_{\max}}{I_{G,\max}} - R_G \dots \dots \dots (3)$$

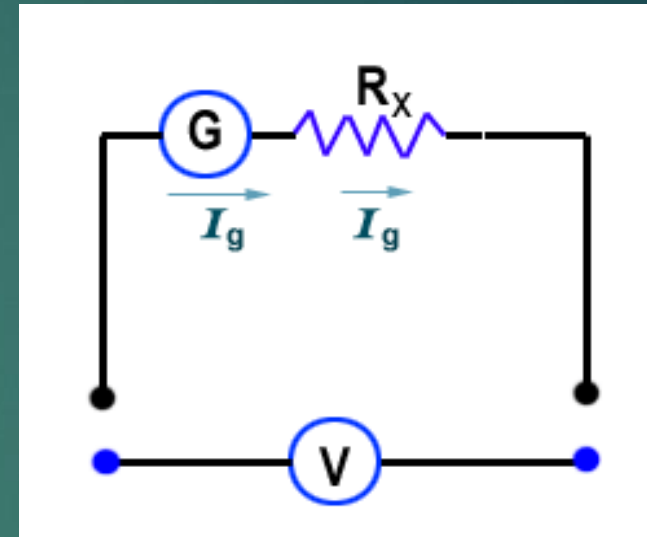


Figure 3

Procedure

1. Calculate the value of the resistance (R_x) from relation (3) to build up an voltmeter measuring a maximum voltage of $V_{\max} = 5\text{Volt}$.
2. Connect the resistance (R_x)-from resistance box- to the galvanometer as shown in figure 3.
3. Connect the circuit shown in the Figure 4
4. By the rheostat change the voltage in the circuit and record the reading of the voltmeter (V) and the built voltmeter (V_d) and record the readings in table 2.
5. Drew the relation between the voltage V and V_d .
6. From the slope calculate the efficiency of the built voltmeter

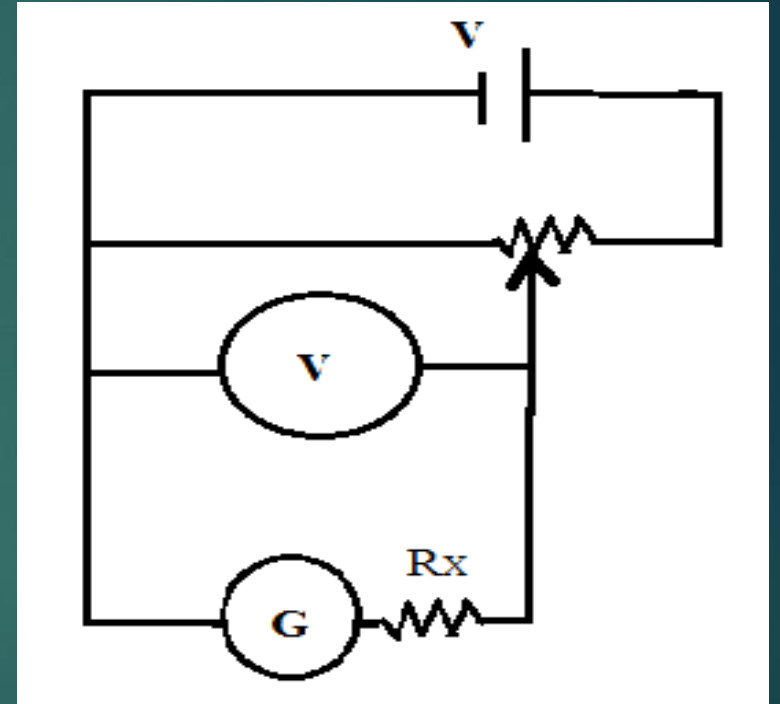


Figure 4



THANK YOU

